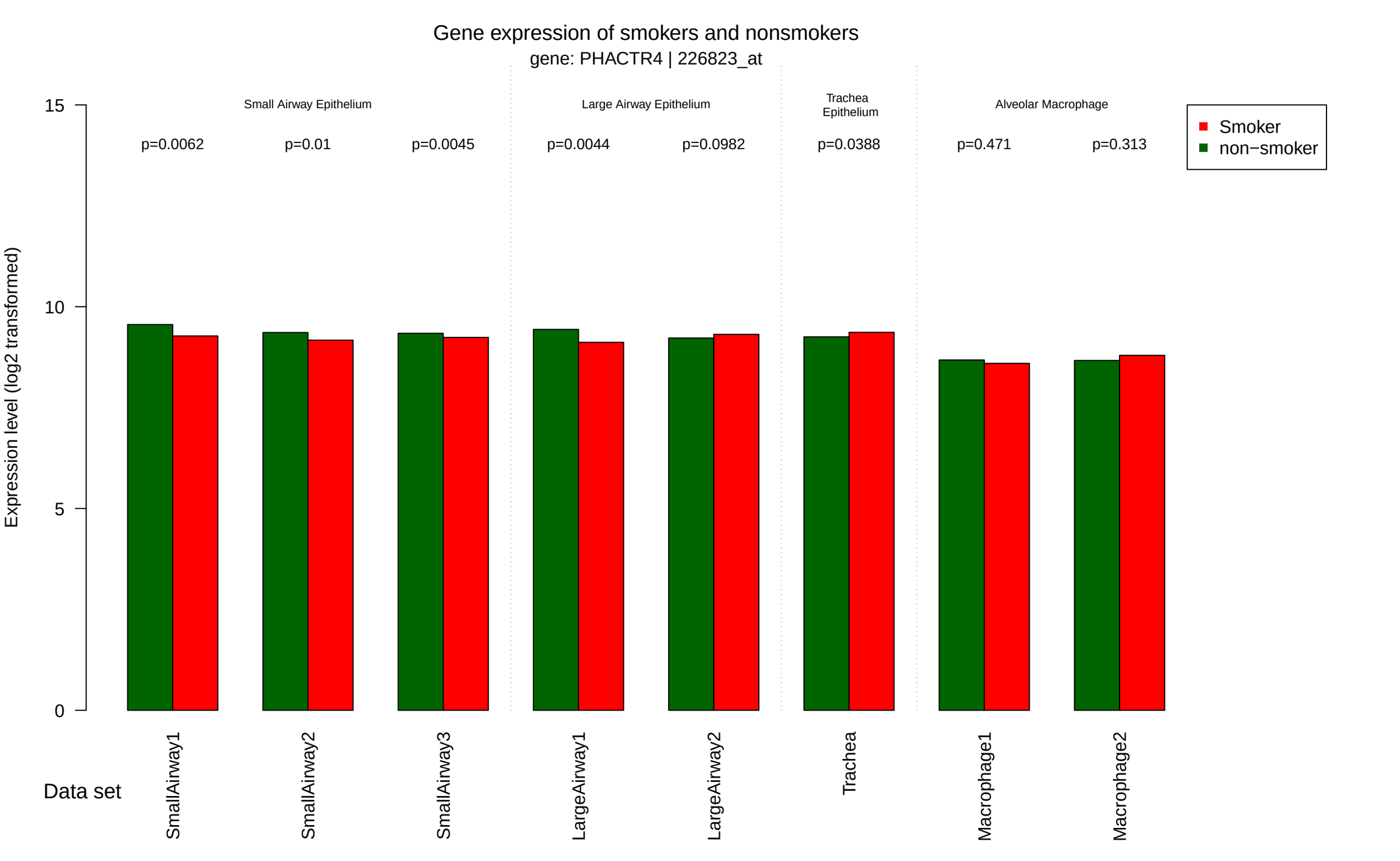
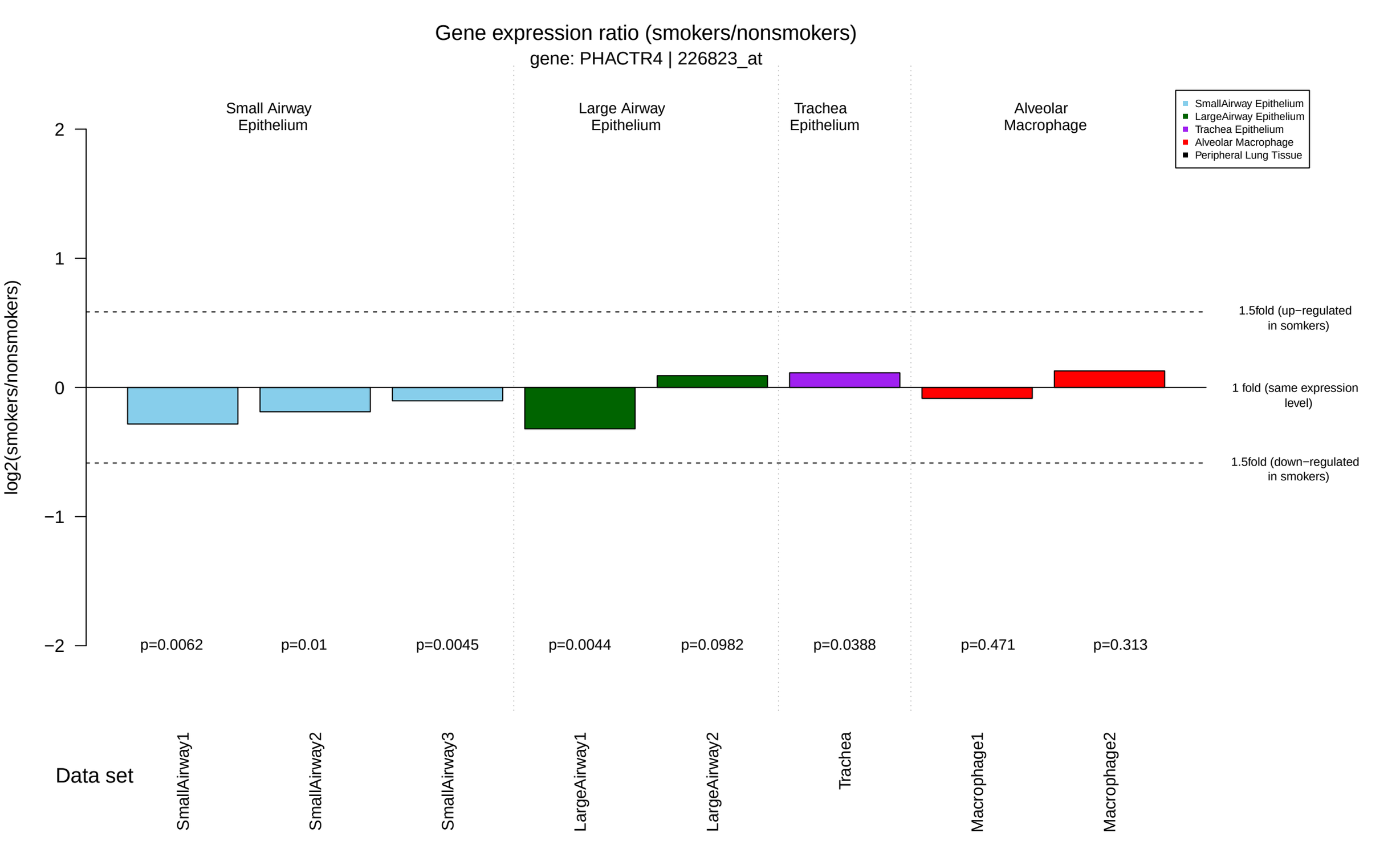
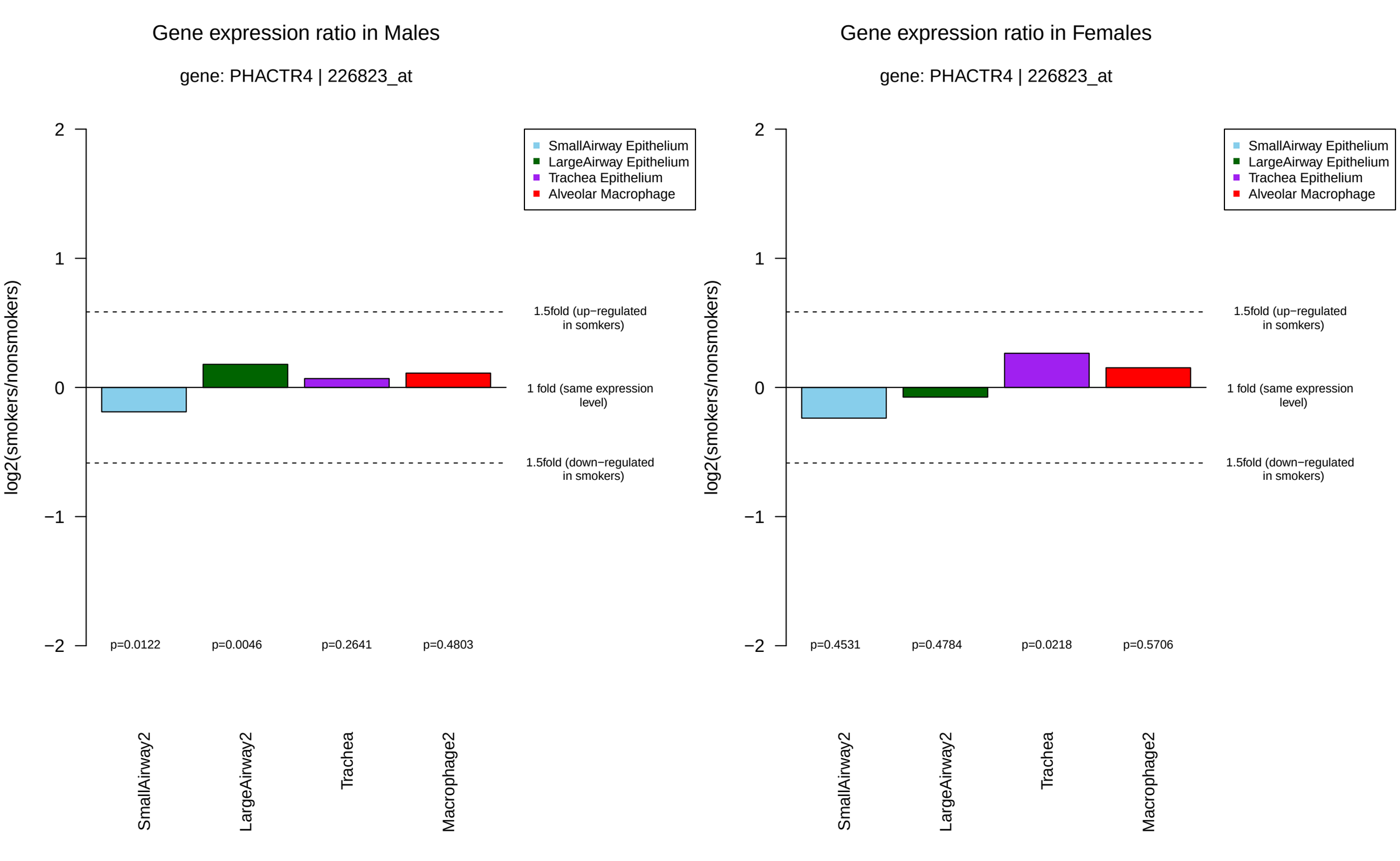
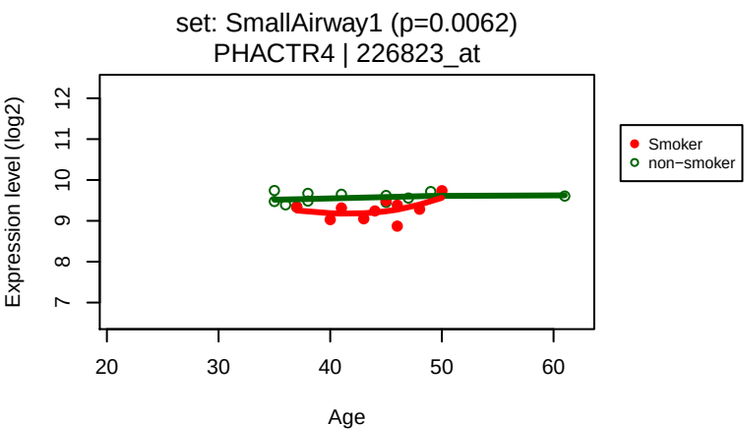
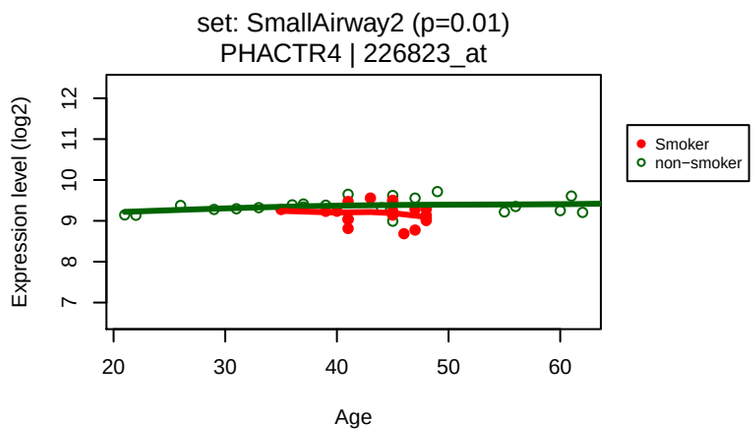
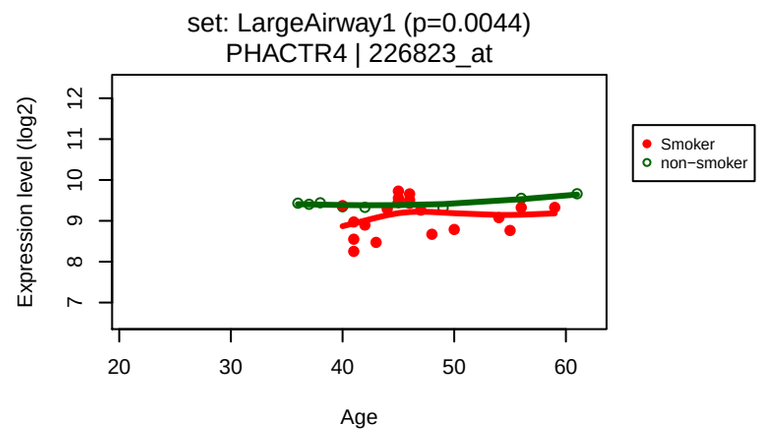

Gene expression of smokers and nonsmokers
gene: PHACTR4 | 226823_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0062
p=0.01
p=0.0045
p=0.0044
p=0.0982
p=0.0388
p=0.471
p=0.313
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

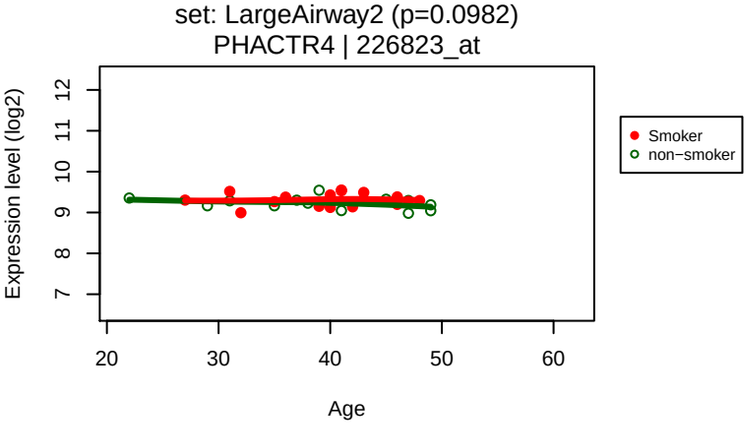

Gene expression ratio (smokers/nonsmokers)
gene: PHACTR4 | 226823_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0062
p=0.01
p=0.0045
p=0.0044
p=0.0982
p=0.0388
p=0.471
p=0.313
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

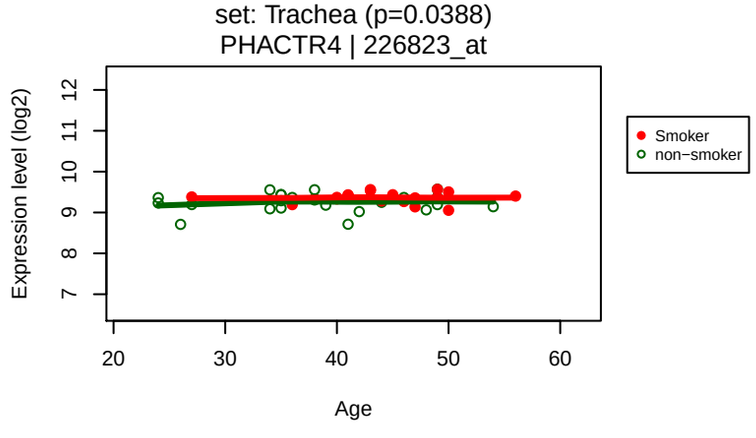

Gene expression ratio in Males
gene: PHACTR4 | 226823_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0122
p=0.0046
p=0.2641
p=0.4803
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PHACTR4 | 226823_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4531
p=0.4784
p=0.0218
p=0.5706
SmallAirway2
LargeAirway2
Trachea
Macrophage2

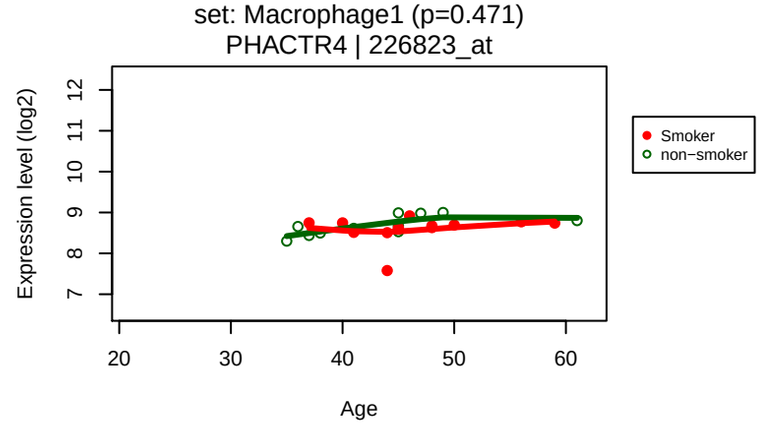

set: SmallAirway1 (p=0.0062)
PHACTR4 | 226823_at
Expression level (log2)
Age
Smoker
non−smoker

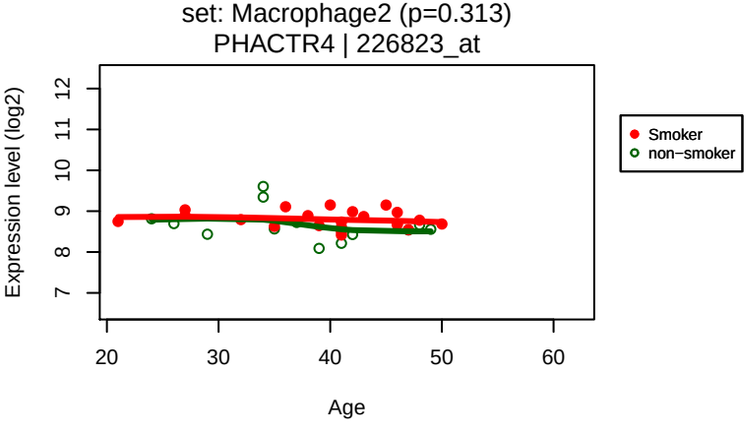

set: SmallAirway2 (p=0.01)
PHACTR4 | 226823_at
Expression level (log2)
Age
Smoker
non−smoker

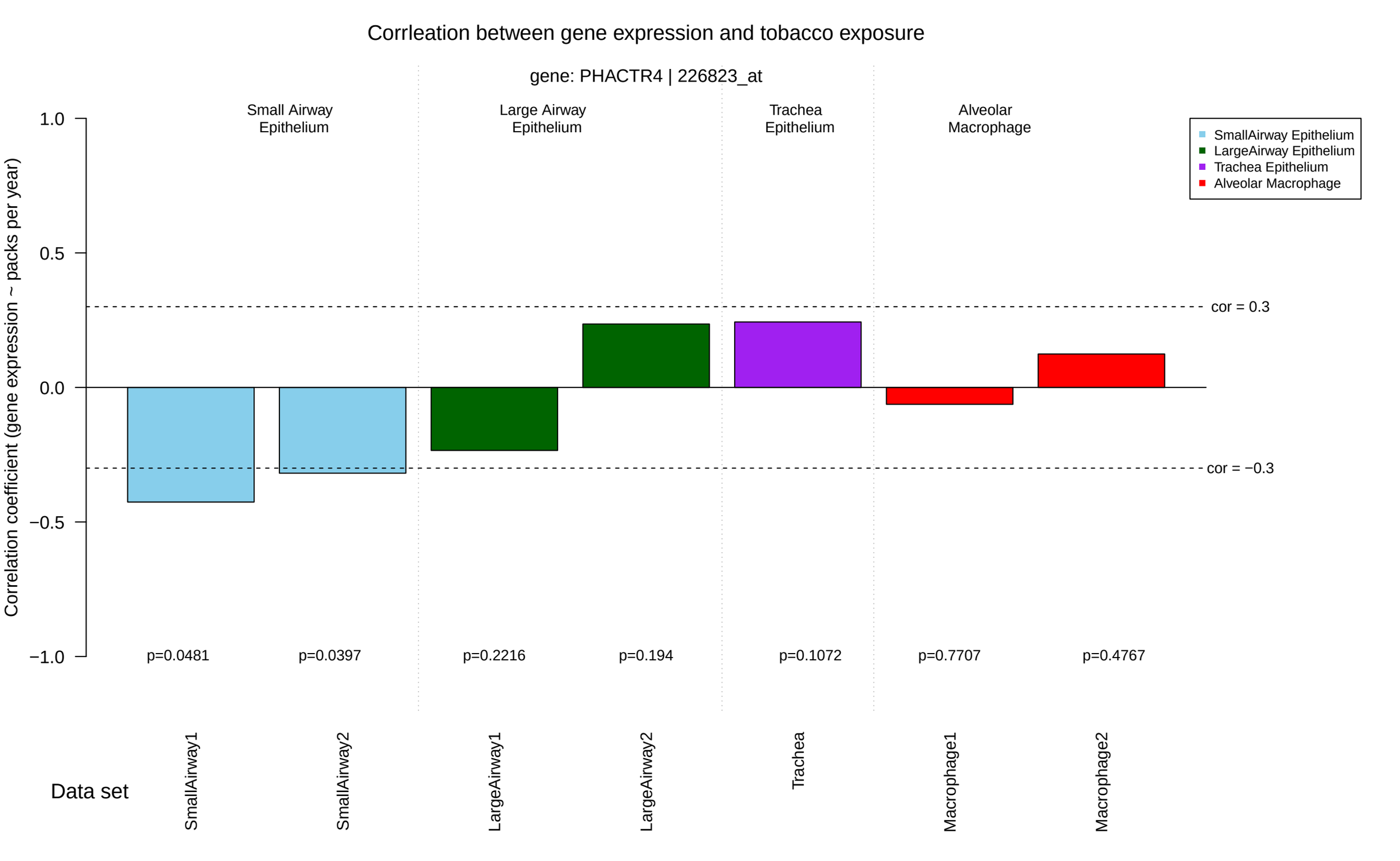

set: LargeAirway1 (p=0.0044)
PHACTR4 | 226823_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0982)
PHACTR4 | 226823_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0388)
PHACTR4 | 226823_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.471)
PHACTR4 | 226823_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.313)
PHACTR4 | 226823_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PHACTR4 | 226823_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.0481
p=0.0397
p=0.2216
p=0.194
p=0.1072
p=0.7707
p=0.4767
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2